

Problem Statement

AI-Assisted Cross-Layer Threat Detection & Explanation System

1. Background & Motivation

Modern cyberattacks rarely stay confined to a single layer of an organization's infrastructure. A typical multi-stage attack might involve unusual network traffic (reconnaissance or exfiltration), suspicious operating system activity (privilege escalation, process injection), and anomalous cloud access patterns (compromised credentials, unusual IAM permission changes) — often occurring within the same intrusion but detected, if at all, by separate, siloed tools.

Modern Security Operations Centers (SOCs) rely heavily on layer-specific tools: network intrusion detection systems (NIDS), endpoint detection and response (EDR), and cloud security posture management (CSPM) platforms. Each generates its own alerts, but correlating those alerts into a single coherent picture of an attack is still largely a manual, expertise-dependent process. This contributes directly to alert fatigue and delayed incident response — two of the most cited pain points in current industry security reports. The security industry is actively moving toward "autonomous SOC" and "extended detection and response" (XDR) models that attempt to solve exactly this problem, making this both a timely and practically relevant research direction.

2. Problem Statement

There is no lightweight, extensible system that (a) detects anomalies independently across network, OS, and cloud layers using machine learning, (b) correlates those anomalies into a single multi-stage attack narrative, and (c) uses a large language model to translate that correlated evidence into a clear, prioritized, human-readable explanation for a security analyst — while also being evaluated for its own robustness against adversarial evasion at each layer.

Most existing academic work addresses detection at a single layer in isolation, or proposes LLM-based security assistants without rigorously testing whether the underlying detection pipeline can itself be evaded. This leaves a gap between "detection accuracy on a clean dataset" and "detection reliability against a motivated attacker" — a gap that matters far more in practice than in a benchmark.

3. Research Gap

- Cross-layer correlation of security signals is addressed more in industry tooling than in reproducible academic research.
- LLM-based alert summarization/explanation for SOC analysts is an emerging area with limited adversarial evaluation.
- Few studies stress-test the full detection pipeline (all layers plus the LLM explanation layer) as a single adversarial target, rather than evaluating each component separately.

4. Objectives

1. Design and implement independent ML-based anomaly detectors for network traffic, OS/endpoint logs, and cloud access logs.
2. Build a fusion mechanism that correlates anomalies flagged across the three layers into unified multi-stage attack scenarios.
3. Integrate an LLM layer that converts correlated technical evidence into a prioritized, human-readable analyst alert.
4. Design realistic multi-stage attack scenarios spanning all three layers and use them to red-team the complete pipeline, including adversarial evasion attempts at each detection layer and prompt-level attacks on the LLM explanation layer.
5. Evaluate detection performance and robustness before and after hardening, with quantitative before/after metrics.

5. Proposed Methodology

- Data: Use publicly available or simulated datasets for network traffic (e.g., NIDS benchmark datasets), OS/endpoint logs, and cloud access logs (e.g., simulated CloudTrail-style logs), combined into synthetic multi-stage attack chains where realistic public data is unavailable.
- Detection layer: Train separate ML models per layer (e.g., classical ML or lightweight deep learning, depending on data volume) to flag layer-specific anomalies.
- Fusion layer: Correlate flagged anomalies across layers using timestamp/entity-based linking (e.g., same user, same host, same time window) to reconstruct attack chains.
- Explanation layer: Use an LLM to generate a natural-language summary and severity ranking from the correlated evidence.
- Adversarial evaluation: Attempt evasion attacks against each detector (e.g., traffic padding, log obfuscation, credential-use mimicry) and prompt-based attacks against the LLM explanation layer; measure detection/explanation degradation before and after applying defenses.

6. Novelty / Contribution

The core contribution is not a new detection algorithm in isolation, but a reproducible cross-layer pipeline combined with an adversarial stress-test of the entire system — including the LLM explanation layer, which is rarely evaluated adversarially in existing work. The before/after evaluation (undefended vs. hardened pipeline) provides a concrete, measurable result suitable for a conference-length paper.

7. Expected Outcome

A working prototype demonstrating cross-layer attack detection and explanation, with a quantitative comparison of detection/explanation reliability before and after adversarial hardening — forming the core results section of the paper.